

Body Mass Index and Early Kidney Function Decline in Young Adults: A Longitudinal Analysis of the CARDIA (Coronary Artery Risk Development in Young Adults) Study

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Background: Identifying potentially modifiable risk factors is critically important for reducing the burden of chronic kidney disease. We sought to examine the association of body mass index (BMI) with kidney function decline in a cohort of young adults with preserved glomerular filtration at baseline.

Study Design: Longitudinal cohort.

Setting & Participants: 2,839 black and white young adults with cystatin C–based estimated glomerular filtration rate (eGFR_{cys}) > 90 mL/min/1.73 m² taking part in the year-10 examination (in 1995-1996) of the Coronary Artery Risk Development in Young Adults (CARDIA) Study.

Predictor: BMI, categorized as 18.5-24.9 (reference), 25.0-29.9, 30.0-39.9, and ≥40.0 kg/m².

Outcomes: Trajectory of kidney function decline, rapid decline (>3% per year), and incident eGFR_{cys} < 60 mL/min/1.73 m² over 10 years of follow-up.

Measurements: GFR_{cys} estimated from the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) equation for calibrated cystatin C at CARDIA years 10, 15, and 20.

Results: At year 10, participants had a mean age of 35.1 years, median eGFR_{cys} of 114 mL/min/1.73 m², and 24.5% had BMI ≥ 30.0 kg/m². After age 30 years, average eGFR_{cys} was progressively lower with each increment in BMI after adjustment for baseline age, race, sex, hyperlipidemia, smoking status, and physical activity. Higher BMI category was associated with successively higher odds of rapid decline (for 25.0-29.9, 30.0-39.9, and ≥40.0 kg/m², adjusted ORs were 1.50 [95% CI, 1.21-1.87], 2.01 [95% CI, 1.57-2.87], and 2.57 [95% CI, 1.67-3.94], respectively). 18 participants (0.6%) had incident eGFR_{cys} < 60 mL/min/1.73 m². In unadjusted analysis, higher BMI category was associated with incident eGFR_{cys} < 60 mL/min/1.73 m² (for 25.0-29.9, 30.0-39.9, and ≥40.0 kg/m², ORs were 5.17 [95% CI, 1.10-25.38], 7.44 [95% CI, 1.54-35.95], and 5.55 [95% CI, 0.50-61.81], respectively); adjusted associations were no longer significant.

Limitations: Inability to describe kidney function before differences by BMI category were already evident. Absence of data for measured GFR or GFR estimated from serum creatinine level.

Conclusions: Higher BMI categories are associated with greater declines in kidney function in a cohort of young adults with preserved GFR at baseline. Clinicians should vigilantly monitor overweight and obese patients for evidence of early kidney function decline.

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INDEX WORDS: Kidney function decline; obesity; risk factor.

Chronic kidney disease (CKD) affects an estimated 14% of US adults and is associated with significant morbidity and mortality.¹ Identification of potentially modifiable risk factors for CKD is essential for reducing its burden.

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Although several longitudinal studies have demonstrated an association between obesity and incident CKD, these observations have been made primarily in cohorts of older individuals²⁻⁴ or among individuals with established kidney disease.^{2,4} Because obesity affects an estimated 17% of children and adolescents, nearly triple its prevalence in the 1980s,^{5,6} determining the association of obesity with declining kidney function in younger populations, as well as understanding the association of obesity on the continuum of incipient kidney disease throughout the life course, is critical.

Within a cohort of young adults with preserved glomerular filtration at baseline, we examined the association of body mass index (BMI) categories with kidney function decline as measured by cystatin C, an alternative biomarker used to estimate glomerular filtration rate (GFR) and thought to be particularly useful for detecting reductions in kidney function at

earlier stages (estimated GFR [eGFR] > 60 mL/min/1.73 m²).⁷ We hypothesized that individuals with higher BMI would have a faster decline in kidney function and more progression to eGFR < 60 mL/min/1.73 m² than their counterparts with normal BMI.

METHODS

Study Design and Population

We conducted a longitudinal analysis of CARDIA (Coronary Artery Risk Development in Young Adults) Study participants. CARDIA is a multicenter cohort study to evaluate the development and determinants of cardiovascular risk factors and disease in young adults. From 1985 to 1986, a total of 5,115 asymptomatic young adults aged 18–30 years were recruited from 4 US cities (Birmingham, AL; Chicago, IL; Minneapolis, MN; and Oakland, CA). The study design, protocol, and recruitment process have been described previously in detail.⁸ The institutional review boards at each site approved the examination protocol, and written informed consent was obtained at every examination. The cohort was designed for balance by race (black and white), sex, age, and education. There were 52% black participants, 55% women, and 40% with 12 or fewer years of formal education. Follow-up examinations were completed at study years 2, 5, 7, 10, 15, 20, and 25. The retention rate of the surviving cohort was 79% at year 10, 74% at year 15, and 72% at year 20. All measurements were obtained for all available participants at every study visit except for cystatin C, which was measured as part of an ancillary study to CARDIA on all available stored plasma samples from participants attending the year-10, -15, and -20 examinations.

Of 3,944 participants who completed the year-10 study visit, we excluded 58 without a year-10 BMI measurement, 76 with BMI < 18.5 kg/m² at year 10, and 118 without a cystatin C measurement at year 10. We excluded 295 participants with eGFR estimated from cystatin C level (eGFR_{cys}) < 90 mL/min/1.73 m² at year 10 in order to examine the cohort with preserved kidney function at baseline (year 10). We excluded 558 additional participants with no year-15 cystatin C measurement because consecutive kidney function measurements were required for our rapid eGFR_{cys} decline and incident CKD outcomes. The participants excluded because of missing year-15 cystatin C values had similar BMI subgroup prevalences compared with the 2,839 participants with available data who were included in this study ($P = 0.09$).

Predictor

Our primary predictor was BMI (in kilograms per meter squared), which we categorized as 18.5–24.9 (reference, normal weight), 25.0–29.9 (overweight), 30.0–39.9 (obese), or ≥ 40.0 kg/m² (extremely obese).⁹ BMI was calculated from body weight measured with light clothing to the nearest 0.2 lb and body height without shoes to the nearest 0.5 cm.

Main Outcome

Our outcome of interest was kidney function decline, which we evaluated in 3 ways: (1) trajectory of eGFR_{cys} (in milliliters per minute per 1.73 m²) using repeated eGFR_{cys} measurements at study years 10, 15, and 20; (2) rapid eGFR_{cys} decline ($> 3\%$ per year in the intervals between years 10, 15, and 20); and (3) incident eGFR_{cys} < 60 mL/min/1.73 m² at study years 15 or 20. Our definition of rapid decline represents a magnitude of change 3 times the expected rate previously described in aging population studies¹⁰ and has been used in several studies.^{11–13} Serum creatinine also was measured at all study years for all available participants; however, as in prior CARDIA Study analyses, we estimated kidney function using cystatin C level because it has

been shown to perform better in evaluating kidney function for persons at higher eGFR ranges^{7,14,15} and may detect changes in kidney function earlier than creatinine level.¹⁶ Furthermore, creatinine values in the CARDIA Study have been measured using different assays over time, thus rendering examination of trajectories in kidney function uninterpretable.

Cystatin C was measured as part of an ancillary study on all stored frozen plasma from study years 10, 15, and 20 by nephelometry using the N Latex Cystatin C kit (Dade Behring) and later calibrated to most recent cystatin C standardization. Kidney function (eGFR in milliliters per minute per 1.73 m²) then was estimated by the 2012 CKD-EPI (CKD Epidemiology Collaboration) cystatin C equation, which is for use with calibrated cystatin C.¹⁷

Covariates

Age, race, sex, smoking status, and hyperlipidemia were defined using year-10 data and were measured on all available CARDIA participants. Age, race (white or black), sex, and smoking status were obtained by self-report. We defined smoking status as never, past, or current use. Hyperlipidemia was defined by low-density lipoprotein cholesterol (derived by the Friedewald equation)¹⁸ level > 130 mg/dL or self-reported lipid-lowering medication use.

We defined diabetes as fasting serum glucose level ≥ 126 mg/dL or self-reported hypoglycemic medication use using data collected at the year-10, -15, and -20 examinations. Serum glucose was measured using hexokinase coupled to glucose-6-phosphate dehydrogenase by Linco Research (St Louis, MO). Three seated systolic and diastolic blood pressure measurements were performed with a random-zero sphygmomanometer. We used the mean of the second and third readings at the year-10, -15, and -20 examinations and considered systolic and diastolic blood pressures separately as continuous variables.

Albuminuria was determined from a single untimed (spot) urine sample collected at the year-10, -15 and -20 examinations.¹⁹ Urine albumin was measured using a nephelometric procedure with a specific anti-albumin monoclonal antibody, and creatinine was assessed using the Jaffé method.¹⁹ Urine albumin-creatinine ratios were standardized to sex and race and expressed in milligrams per gram of creatinine.²⁰ Albuminuria was defined as urine albumin-creatinine ratio > 30 mg/g.

Serum high-sensitivity C-reactive protein (CRP; in milligrams per liter) was measured at the year-7, -15, and -20 examinations with a BNII nephelometer (Dade Behring). Because CRP was not measured at year 10, we used the year-7 data as the baseline measurement. Because values were not normally distributed, we log₂ transformed them to downweight noisier values in the higher ranges.

Physical activity was defined by an interviewer-administered questionnaire at each CARDIA examination. This validated questionnaire asked about frequency of participation in 13 categories of moderate and vigorous recreational sports, exercise, leisure, and occupational activities during the previous 12 months.²¹ We used log-transformed physical activity scores, which were calculated in exercise units based on frequency and intensity of each activity.

Statistical Analyses

We estimated the trajectory of population mean eGFR_{cys} values across the entire age spectrum from 30–50 years, using linear mixed models in which age effects were captured using 5-knot restricted cubic splines, specific to each BMI group. Individual departures from the trajectory of the population mean were modeled using random intercepts and cubic spline components. We examined population mean eGFR_{cys} graphically by age for each BMI group separately, which included a total of 8,146 eGFR_{cys} measurements over 26,535 person-years. We tested for differences in slopes across BMI categories of eGFR_{cys} trajectories

Table 1. Baseline Participant Characteristics by BMI Group

	BMI 18.5-24.9 (n = 1,207)	BMI 25.0-29.9 (n = 936)	BMI 30.0-39.9 (n = 583)	BMI ≥ 40 (n = 113)	P ^a
eGFR _{cys} (mL/min/1.73 m ²)	115 [109-120]	114 [107-119]	112 [105-118]	110 [100-114]	<0.001
Age (y)	35.0 ± 3.6	35.2 ± 3.6	35.1 ± 3.7	35.02 ± 3.9	0.7
Race and sex					<0.001
Black women	213 (17.7)	214 (22.9)	249 (42.7)	71 (62.8)	
Black men	189 (15.7)	214 (22.9)	131 (22.5)	15 (13.3)	
White women	491 (40.7)	184 (19.7)	101 (17.3)	17 (15.0)	
White men	314 (26.0)	324 (34.6)	102 (17.5)	10 (8.9)	
Smoking status					0.03
Never	503 (41.7)	416 (44.4)	282 (48.4)	61 (54.0)	
Past	424 (35.1)	300 (32.0)	164 (28.2)	31 (27.4)	
Current	280 (23.2)	220 (23.5)	136 (23.4)	21 (18.6)	
Hyperlipidemia ^b	183 (15.2)	261 (27.9)	176 (30.2)	31 (27.4)	<0.001
Physical activity (exercise units)	5.7 [5.0-6.2]	5.7 [5.0-6.2]	5.3 [4.5-5.9]	5.3 [4.4-5.9]	<0.001
Diabetes ^c	41 (3.4)	38 (4.1)	69 (11.8)	26 (23.0)	<0.001
Systolic BP	106.1 ± 11.2	110.6 ± 11.3	114.1 ± 12.8	116.4 ± 14.4	<0.001
Diastolic BP	69.7 ± 9.0	72.9 ± 9.4	75.9 ± 10.2	77.2 ± 9.3	<0.001
C-Reactive protein (mg/L)	0.65 [0.30-1.58]	0.96 [0.51-2.3]	2.60 [1.20-4.80]	6.76 [3.97-10.20]	<0.001
Albuminuria > 30 mg/g	31 (2.9)	24 (2.8)	32 (6.1)	10 (10.2)	<0.001

Note: N = 2,839. Baseline indicates CARDIA year 10 (except baseline C-reactive protein, which was measured at CARDIA year 7 but not year 10). BMI expressed in kg/m². Values for categorical variables are given as number (percentage); values for continuous variables are given as mean ± standard deviation or median [interquartile range].

Abbreviations: BMI, body mass index; BP, blood pressure; CARDIA, Coronary Artery Risk Development in Young Adults; eGFR_{cys}, serum cystatin C–based estimated glomerular filtration rate.

^aP value is analysis of variance (age, systolic BP, diastolic BP), Wilcoxon (eGFR_{cys}, physical activity, C-reactive protein), or χ^2 *t* test (all other) of association between BMI category and characteristic.

^bDefined as low-density lipoprotein cholesterol > 130 mg/dL or receiving lipid medication.

^cDefined as fasting serum glucose > 126 mg/dL or receiving diabetes medication.

adjusting for baseline age (to control for possible cohort effects), race, sex, smoking status, hyperlipidemia, and physical activity as potential confounders. Because physical activity was defined by repeated measures, it was considered as a time-dependent covariate in all models.

We used repeated-measures logistic regression with robust standard errors to examine associations of BMI categories with rapid eGFR_{cys} decline (>3% per year) in the intervals between years 10, 15, and 20 and a pooled logistic model to assess associations with incident eGFR_{cys} < 60 mL/min/1.73 m² at year 15 or 20, adjusting for age, race, sex, smoking status, hyperlipidemia, and physical activity (multivariate model). Because cystatin C is produced by all nucleated cells, including adipose cells, we performed a sensitivity analysis for each outcome in which we additionally adjusted multivariate models for changes in BMI over time. Because participants with no year-15 cystatin C value were excluded to examine BMI association with rapid eGFR decline and incident eGFR < 60 mL/min/1.73 m², we performed an additional sensitivity analysis in which we repeated the analysis for mean eGFR_{cys} trajectory including these participants.

For all outcomes, we then performed a series of analyses on the multivariate model, sequentially adding incident diabetes, blood pressure, CRP level, and albuminuria. These covariates were added individually and then simultaneously to determine whether the associations of BMI with kidney function decline were mediated by these factors. Because incident diabetes, systolic blood pressure, diastolic blood pressure, albuminuria, and CRP level were defined by repeated measures, they were considered as time-dependent covariates in the models. Finally, we tested the multivariate model of each outcome for interactions of BMI with

race and sex. Analyses were performed using SAS, version 9.2 (SAS Institute Inc), and Stata, version 13 (StataCorp LP).

RESULTS

Mean BMI in the study sample at baseline (CARDIA year 10) was 27.2 ± 5.9 (SD) kg/m², and 33.0%, 20.5%, and 4.0% had BMI of 25.0-29.9, 30.0-39.9, and ≥40.0 kg/m², respectively. Median eGFR_{cys} was 114 (interquartile range [IQR], 107-119) at baseline, 112 (IQR, 105-117) at year 15, and 106 (IQR, 96-112) mL/min/1.73 m² at year 20 among participants overall. Participant characteristics by BMI category are shown in Table 1. Mean age of participants included in the study was 35.1 years at baseline and was not different by BMI group. BMI ≥ 40.0 kg/m² was present in 9.5%, 2.7%, 2.1%, and 1.3% of black women, black men, white men, and white women, respectively. Mean systolic and diastolic blood pressures and prevalence of baseline diabetes, hyperlipidemia, never smoking, and albuminuria were higher in the higher BMI categories. Median CRP level was 10 times as high in the group with BMI ≥ 40 kg/m² compared with BMI of 18.5-24.9 kg/m².

The overall association between BMI category and eGFR_{cys} trajectory is shown in Fig 1. Detailed

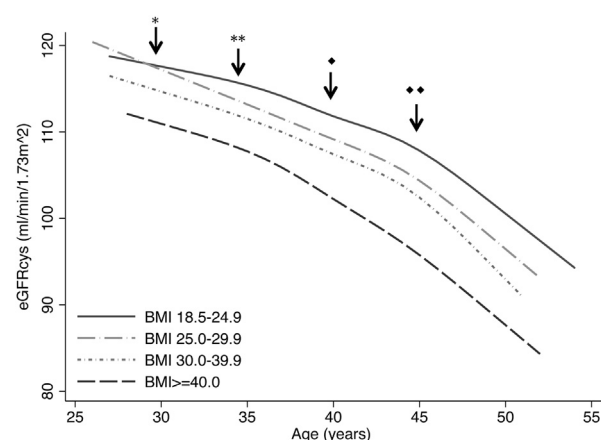


Figure 1. Trajectory of cystatin C–estimated glomerular filtration rate (eGFR_{cys}; in mL/min/1.73 m²) decline (adjusted for baseline age, race, sex, hyperlipidemia, smoking status, and physical activity) by age and body mass index (BMI; in kg/m²) category. Differences in eGFR_{cys} slope: *BMI of 25.0–29.9 versus BMI of 18.5–24.9 kg/m², $P = 0.01$; **BMI of 25.0–29.9 versus BMI of 18.5–24.9 kg/m², $P = 0.002$; (black diamond) BMI ≥ 40 versus BMI of 18.5–24.9 kg/m², $P = 0.02$; (double black diamonds) BMI of 30.0–39.9 versus BMI of 18.5–24.9 kg/m², $P = 0.002$.

numeric results are shown in Table S1 (provided as online supplementary material). After age 30 years, average eGFR_{cys} was progressively lower with each increment of BMI category after adjustment for confounders, as well as potential mediators. Across all ages, the rate of eGFR_{cys} decline was nominally slowest for the group with BMI of 18.5–24.9 kg/m², but differences were not statistically significant for most comparisons. Adjustment for change in BMI over time modestly attenuated between-group differences in

eGFR_{cys} decline (Table S2). Results of sensitivity analysis including participants with no year-15 cystatin C measurement (who were excluded for the rapid eGFR decline and incident eGFR_{cys} < 60 mL/min/1.73 m² outcomes) were not substantively different. Although associations between BMI categories and eGFR_{cys} trajectory varied significantly by sex ($P < 0.001$) and race ($P < 0.001$), BMI ≥ 25.0 kg/m² was associated with faster eGFR_{cys} decline in all 4 race/sex subgroups (Fig S1).

Overall, rapid eGFR_{cys} decline ($>3\%$ per year) was observed in 569 (10.7%) of the 5,307 five-year person-intervals included in the analysis. Results of the primary analyses for the rapid eGFR_{cys} decline outcome are shown in Table 2. Relative to the referent group with BMI of 18.5–24.9 kg/m², each BMI category was associated significantly with successively higher likelihoods of rapid eGFR_{cys} decline in the unadjusted and confounder-adjusted models. Additional adjustments for changes in BMI over time, incident diabetes, albuminuria, and CRP level did not change the association appreciably. There was evidence that the association between BMI categories and rapid eGFR_{cys} decline was significantly stronger in women than men (P for interaction = 0.005) and somewhat stronger in whites than blacks (P for interaction = 0.1; Table 3).

Only 18 (0.6%) participants developed incident eGFR_{cys} < 60 mL/min/1.73 m² during the 10-year study period. Compared to the 18.5–24.9 kg/m² reference group, in the unadjusted model, higher BMI category was associated with an increased likelihood of incident eGFR_{cys} < 60 mL/min/1.73 m² for the

Table 2. Odds of Rapid eGFR_{cys} Decline by BMI Category

Model	BMI 25.0–29.9 (203/1,713 [11.5%])		BMI 30.0–39.9 (153/1,078 [14.2%])		BMI ≥ 40.0 (34/208 [16.3%])	
	OR (95% CI)	P	OR (95% CI)	P	OR (95% CI)	P
Unadjusted	1.53 (1.24–1.89)	<0.001	1.98 (1.57–2.49)	<0.001	2.36 (1.57–3.56)	<0.001
Adjusted MV model ^a	1.50 (1.21–1.87)	<0.001	2.01 (1.57–2.87)	<0.001	2.57 (1.67–3.94)	<0.001
MV model + BMI change	1.49 (1.19–1.87)	<0.001	2.00 (1.56–2.58)	<0.001	2.74 (1.79–4.19)	<0.001
MV model + incident diabetes	1.51 (1.21–1.88)	<0.001	2.05 (1.60–2.64)	<0.001	2.65 (1.71–4.11)	<0.001
MV model + repeated systolic and diastolic BP measurements	1.44 (1.15–1.79)	0.001	1.85 (1.44–2.37)	<0.001	2.33 (1.58–3.54)	<0.001
MV model + repeated CRP measures	1.48 (1.17–1.86)	0.001	1.96 (1.48–2.59)	<0.001	2.46 (1.52–4.00)	<0.001
MV model + repeated albuminuria measures	1.56 (1.25–1.96)	<0.001	2.02 (1.57–2.61)	<0.001	2.36 (1.51–3.70)	<0.001
Fully adjusted (includes confounders + all potential mediators ^b)	1.50 (1.18–1.91)	0.001	1.95 (1.45–2.61)	<0.001	2.27 (1.36–3.78)	0.002

Note: Rapid eGFR_{cys} decline defined as $>3\%$ decline per year. BMI expressed in kg/m²; column headings also give n/N [%], which corresponds to number of events per number of 5-year person-intervals, with the corresponding percentage. Reference group is BMI of 18.5–24.9 kg/m² (179/2,263 [7.9%]).

Abbreviations: BMI, body mass index; BP, blood pressure; CI, confidence interval; CRP, C-reactive protein; eGFR_{cys}, serum cystatin C–based estimated glomerular filtration rate; MV, multivariate; OR, odds ratio.

^aAdjusted for potential confounding by age, race, sex, hyperlipidemia, smoking status, and physical activity.

^bPotential mediators include incident diabetes, interim systolic and diastolic BP measurements, repeated CRP level, and repeated albuminuria measures.

Table 3. Odds of Rapid eGFR_{cys} Decline by BMI Category, Stratified by Race and Sex

Subgroup	BMI 18.5-24.9			BMI 25.0-29.9			BMI 30.0-39.9			BMI ≥ 40.0			P interaction ^c
	Events ^a	OR (95% CI)	P ^b	Events ^a	OR (95% CI)	P ^b	Events ^a	OR (95% CI)	P	Events ^a	OR (95% CI)	P	
Men	96/924 (10.4%)	1.00 (reference)		119/973 (12.2%)	1.39 (0.92-1.63)	0.2	56/410 (13.7%)	1.45 (1.01-2.07)	0.04	9/44 (20.4%)	2.27 (1.09-4.73)	0.03	0.005
Women	83/1,339 (6.2%)	1.00 (reference)		78/740 (10.5%)	1.88 (1.34-2.63)	<0.001	90/628 (14.3%)	2.74 (1.92-3.92)	<0.001	22/152 (14.5%)	3.06 (1.76-5.33)	<0.001	
Blacks	63/724 (8.7%)	1.00 (reference)		77/788 (9.8%)	1.17 (0.81-1.68)	0.4	84/669 (12.6%)	1.62 (1.13-2.31)	0.009	23/150 (15.3%)	2.28 (1.33-3.89)	0.004	0.1
Whites	116/1,539 (7.5%)	1.00 (reference)		120/925 (13.0%)	1.74 (1.31-2.31)	<0.001	62/369 (16.8%)	2.52 (1.78-3.57)	<0.001	8/46 (17.4%)	2.66 (1.18-5.97)	0.02	

Note: Rapid eGFR_{cys} decline defined as >3% decline per year (adjusted for age, race, sex, hyperlipidemia, smoking status, and physical activity). BMI expressed in kg/m².

Abbreviations: BMI, body mass index; CI, confidence interval; eGFR_{cys}, serum cystatin C–based estimated glomerular filtration rate; OR, odds ratio.

^aNumber of events per number of 5-year person-intervals, with the corresponding percentage, within subgroup.

^bP value for difference in rate of decline compared to reference BMI category.

^cP for interaction for BMI category and subgroup.

groups with BMI of 25.0-29.9 (odds ratio [OR], 5.17; 95% confidence interval [CI], 1.10-25.38; $P = 0.04$) and 30.0-39.9 kg/m² (OR, 7.44; 95% CI, 1.54-35.95; $P = 0.01$), but did not reach statistical significance for the group with BMI ≥ 40.0 kg/m² (OR, 5.55; 95% CI, 0.50-61.81; $P = 0.2$). These findings were similar after adjusting for age, race, sex, hyperlipidemia, smoking status, and physical activity, although the association remained statistically significant for only the group with BMI of 30.0-39.9 kg/m² (adjusted OR, 6.10; 95% CI, 1.08-34.41; $P = 0.04$). The small number of outcomes prevented testing of interactions between BMI category and race and sex.

DISCUSSION

In a cohort of black and white young adults free of kidney disease at baseline, we found that increasing BMI was associated significantly with lower kidney function throughout the 10-year study period, with some evidence of faster kidney function decline throughout the age spectrum. Furthermore, we found that higher BMI categories were associated strongly with a higher likelihood of rapid kidney function decline, even after adjustment for important confounders and potential mediating factors.

Prior investigation of participants in the Framingham Offspring Study who had a mean age of 43 years and were free of kidney disease at baseline found that each standard deviation increase in BMI was associated with an ~20% increased likelihood of incident CKD during a mean 18.5-year follow-up.³ Our study builds on our understanding of the potential effect of obesity over the life course of kidney function by examining younger participants and by additionally examining the trajectory of kidney function decline and rapid kidney function decline. By using cystatin C level to estimate GFR, we are able to describe declines in kidney function that may occur within the normal GFR range. We confirmed the importance of BMI as a risk factor for declining kidney function and determined that the association increases with successively higher BMI from 25.0 kg/m², can be observed in early adulthood, and is related to rapid decline in eGFR within the normal GFR range. Furthermore, we have extended understanding in this area by examining a cohort comprising blacks and whites and confirmed that the association between BMI and kidney function is observed for both men and women and blacks and whites.

To our knowledge, only 2 prior studies have examined the association between obesity and kidney function in a younger cohort. Vivante et al²² found that obesity was associated with a 3-fold increased risk of incident end-stage renal disease over 25 years in a cohort of Israeli Army volunteers who were 17 years old at baseline. In a British cohort,

Silverwood et al²³ found that those who became overweight as defined by BMI $> 25.0 \text{ kg/m}^2$ at age 26 or 36 years were approximately twice as likely to have CKD (decreased eGFR and abnormal albuminuria) at age 60–64 years than those who never became overweight. However, lacking interim measurements for kidney function or potential mediators, neither study can describe the trajectory of kidney function decline over time or potential pathways for obesity's effect on kidney function decline. An important strength of our study is that we were able to examine interval development of diabetes and hypertension as potential mediators of the association between BMI and kidney function decline.

We also examined albuminuria and inflammation as potential mediators of our findings, given that obesity is thought to lead to direct kidney injury through a glomerulopathy resembling focal segmental glomerulosclerosis with enlarged glomeruli,²⁴ and because obesity also is recognized as a state of low-grade inflammation,²⁵ with several studies^{26–28} implicating inflammatory markers in the progression of kidney disease in older and elderly populations. However, a significant association between BMI and kidney function remained after simultaneously accounting for all potential mediators, suggesting the presence of other mechanisms for the potential negative effect of BMI on kidney function.

We determined that the slope of decline in eGFR may differ significantly in higher BMI categories compared to normal BMI, which may have important implications for monitoring and early interventions in this high-risk population. However, our finding that obese (BMI, 30–39.9 kg/m^2) and extremely obese (BMI $\geq 40.0 \text{ kg/m}^2$) participants in their late 20s already have lower eGFRs than their normal-weight (BMI, 18.5–24.9 kg/m^2) counterparts suggests that obesity may have an important impact on kidney function at even younger ages. Further study of even younger adults and adolescents is needed to inform this question of when increased BMI may begin affecting eGFR. Only one study to date has examined the association between cystatin C level and BMI in adolescents and found that cystatin C level was correlated positively with BMI.²⁹ Because BMI cannot distinguish muscle mass from fat mass, it has been questioned as a less accurate measure of adiposity, and therefore obesity, than waist circumference.³⁰ However, additional adjustment for waist circumference fully attenuated our findings, suggesting significant overlap between the 2 measures and reinforcing that both are good measures of obesity for this analysis. Further study examining waist circumference as a predictor of kidney function decline would be interesting. Regardless, the impact of obesity of kidney function early in life is of critical

importance given the high prevalence of obesity in childhood.

Cystatin C has properties that make it superior to creatinine for estimating GFR because it is not affected by muscle mass, but rather is made by all nucleated cells, including adipose cells. Therefore, some have questioned its accuracy for obese patients.³¹ However, Schuck et al³² found that obesity had no effect on the relationship between cystatin C level and GFR measured by inulin clearance in adults with CKD. Regardless, these concerns are unlikely to affect the overall interpretation of our study findings because we observed trajectories in decline over time for all BMI categories. Further, adjusting for change in BMI over time resulted in an expected modest attenuation of associations, suggesting that ongoing weight gain explains a portion of the observed kidney function decline.

Although replication of our findings using creatinine-based GFR estimating equations would further strengthen confidence in our cystatin C-based findings, we were unable to do so within the CARDIA Study due to changes in creatinine measurement assays over time. Future confirmatory studies using creatinine-based equations are needed.

A progressive decrease in kidney function has a widespread effect on health, including on the functioning of the heart and brain; hormonal balance, anemia, and bone and mineral metabolism; and ability to resist infections.³³ Our finding of independent associations between BMI and kidney function decline in a cohort of young adults with preserved glomerular filtration at study baseline emphasizes a need for effective obesity interventions early in life. Although dramatic weight loss following bariatric surgery may improve GFR,^{34–36} a recent meta-analysis²⁵ found that modest nonsurgical weight loss did not improve creatinine-based estimates of GFR in those with significantly decreased kidney function, suggesting that weight loss may have been too late to improve kidney function or that a threshold effect for weight loss intervention may exist. Further study is needed to determine the effect of various degrees of weight loss on reversing or stabilizing potential obesity-related declines throughout the spectrum of kidney function. Because individual lifestyle modification often is short-lived and prone to relapse,³⁷ evidence of early kidney function decline may help motivate sustained weight loss to prevent deleterious CKD outcomes in overweight and obese individuals.

In conclusion, higher BMI is associated with greater declines in kidney function in a cohort of young adults with preserved glomerular filtration at study baseline. Given the growing epidemic of obesity in adults and children, the long-term impact of these findings could be substantial. Clinicians should

vigilantly monitor overweight and obese patients for evidence of early kidney function decline.

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SUPPLEMENTARY MATERIAL

Table S1: Annual eGFR_{cys} decline, by age and BMI category.

Table S2: Annual eGFR_{cys} decline, by age and BMI category, multivariate model.

Figure S1: Trajectory of eGFR_{cys} decline by age and BMI category.

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